ZOO503 MIDTERM- 100% GUESS PAPER 2023

Zoogeography-and-Paleontology

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OBJECTIVES

1. Modification of the Oparin ---------- Haldane biochemical model is ***(Hydrothermal Model)***
2. Chromistans Includes all of the following, except……***(Protozoa)***
3. Jawless fishes, ostrachoderms were jawless, armored and had heyday in the --------period ***(Devonian)***
4. Reptiles ancestors of mammals are ***(basal synapsids)***
5. Isthmus of panama separated ***(atlantic and pacific ocean)***
6. Modern fauna developed during ***(550mya)***
7. Basal part of metazoan tree is made from ***(cnidarian)***
8. Refractories are ***(hard organic carbon)***
9. There are ***(five)*** man extinction of brachiopods.
10. Octopus has ***(lost skeleton)***
11. Placoderm were ***(oldest jawed)*** fishes
12. George ***Cuvier*** is known as father of comparative anatomy
13. Fossils information help us in identifying the level of ***(thermal)*** maturity of rocks
14. The order and succession of rocks is called ***(lithostratography)***
15. Cretaceous – Tertiary exticiton------ ***(Mesozoic , Cenozoic)***
16. Order and succession of rocks is called...***(lithostratography)***
17. Isthmus of panama separated...***(Atlantic and Pacific ocean)***
18. Oparin halin biochemical model is... ***Hydrothermal***
19. Chromistan includes all except...***(protozoa)***
20. If there is no change in proportions during growth, the feature is said to show ... ***(isometric)***
21. Diatoms have_____ ***(silicic skeletons)***
22. Planula is larval stage of the_____***(cnidarians)***
23. The composition and structure of fossil communities._____***(paleasynecology)***
24. Bowerbankia is a genus of_____ ***(Bryozoans)***
25. Aplacophorans are _____ ***(shell-less mollusks)***
26. Rostroconchs is _____***(small class mollusks, have 35 genra)***
27. Evolution by natural selection model was presented by _____ ***Charles Darwin in 1830s***
28. In _____(1972) true patterns of relationship are constructed------***(1972 Raup)***
29. The _____have pones in body ***(sponges)***
30. The order and succession of rocks is called _____._____***(Lithostratigraphy)***
31. The crustaceous tertiary extinction change the ecosystem so much that it is regarded us the bond between the ***(15KM crater and 10KM Meteorite)*** eras.
32. Fossils in Fold belts are, except…… ***(In Excellent Shape)***
33. Rohde and Muller ***(2005)*** argued that there is____ periodicity in mass extinctions***----(62 myr)***
34. Periodicity in mass extinctions? ---------- ***(Rohde and Muller (2005))***

35. Modification of the Oparin_____Haldane biochemical model is----- ***(Hydrothermal Model).***

36. Chromistans Includes all of the following, except…….._____***(Protozoa)***

37. Chelicerates are diversed and heterogeneous which includes….____ ***{All of the above (Mites, Scorpions, Spiders)}***

38. Water Vascular system is present in…….____***(Echinoderms)***

39. Types of jawfish--------***(4)***

40. Amniotes radiated in……..__***(Late carboniferous)***

41. Isthmus Of Panama separates----------***(Atlantic and Pacific oceans)***

42. Diatoms have_____ ***(silicic skeletons)***

43. Reptiles ancestors of mammals are _____***(basal synapsids).***

44. Archaeopetyx is true missing link between…….____***(Dinosaurs and Birds)***

45. Trilobites dominated assemblages in England and Wales area were described by ___***(William Smith)***

46. Where an organism live matters in which of the filters is preservation of the complete fossil record……***(Ecological Filters)***

47. When local ecological conditions affect the form of an organism during its life time and consequently fossil is called ……..***(Ecophenotypic Effects)***

48. The Phanerozoic and the current extinction crises is sometime called ***(6***th extinction)

49. Modification of the Oparin_____Haldane biochemical model is---------- ***(Hydrothermal Model).***

50. ___Biomarker indicates that eukaryotes were present around at least 2.7 Ga….***(Lipid)***

51. Chromistans Includes all of the following, except……..____***(Protozoa)***

52. Water Vascular system is present in…….____.***(Echinoderms)***

53. Jawless fishes that lived in ----------period ***(Devonian)***

54. Amniotes radiated in……..____***(Late carboniferous)***

55. In early corocodilian were most diverse……..***(Jurassic and Cretaceous)***

56. Isthmus of panama separated---------- ***(atlantic and pacific ocean)***

57. Pyrenes are ranges of the _____ which are located in France and Spain. ***(mountain)***

58. Metazoans evolve after-------***(emergence of single-celled organisms)***

59. Faunas of Phanerozoic, Cambrian, Paleozoic modern Developed during last----- ***(550 myr)***

60. Among invertebrates _____ have complete fossils records-----***(Paddle worms)***

61. Shallow-water coral reefs exist less than -------***18 C^0***

62. Old bryozoans appeared during----------***(cambrain)***

63. The ---------- are called belly- footed mollusks _____***(Gastropods)***

64. Rostroconchs is a _____***(class of mullusks)***

65. The geological time show----------climates zones_____***(five)***

66. Evolution by natural selection model was presented by _____ ***(Charles Darwin in 1830s)***

67. In _____(1972) true patterns of relationship are constructed------***(1972 Raup)***

68. Father of comparative anatomy is _____***(Cuvier)***

69. Cretaceous – Tertiary exticiton------ ***(Mesozoic , Cenozoic).***

70. When local ecological conditions affect the form of an organism during its life time and consequently fossil is called ……..***(Ecophenotypic Effects)***

SUBJECTIVE

When insects were appeared and how many living insects are there?

Insects first appeared during the Early Devonian and diversified rapidly. There are probably 10 million species of living insects. Insects had probably already evolved flight before the Mid Carboniferous, when giant dragonflies patrolled the forests.

What is the Stratigraphic unit:

A stratigraphic unit is a volume of rock of identifiable origin and relative age range that is defined by the distinctive and dominant, paleontologic features. It is fundamental Unit of Lithostratigraphy. Lithostratigraphic units are bodies of rocks, bedded or unbedded.

What are the three membrane enclose the embryo in reptiles and also there function.

The three membranes that enclose the embryo in reptiles are the amnion, chorion, and allantois.

1. The **amnion** provides protection to the developing embryo by surrounding it in a fluid-filled sac. It acts as a cushion against mechanical shocks and external pressure.
2. The **chorion**, the outer membrane, facilitates gas exchange. It allows oxygen to enter the embryo and carbon dioxide to be expelled.
3. The **allantois**, a membranous sac growing from the hindgut, serves multiple functions. It primarily stores and excretes waste products, including nitrogenous wastes like uric acid.

Name of climate zones:

1= Humid Tropical (no winters and average temperatures above 18°C) 2= Dry subtropical (evaporation exceeds precipitation) 3= Warm temperate (mild winters) 4= Cool temperate (severe winters) 5= Polar (no summers and temperatures below 10°C.

Write and explain types of skeleton.

There are three main types of skeletal systems found in the animal kingdom: hydrostatic skeletons, exoskeletons, and endoskeletons.

1= Hydrostatic Skeleton: A hydrostatic skeleton is a fluid-filled cavity surrounded by muscles. It is primarily found in soft-bodied invertebrates such as worms, jellyfish, and cnidarians. **2= Exoskeleton:** An exoskeleton is a hard, external structure that encloses and protects the body. It is made up of a tough substance called chitin. **3= Endoskeleton:** An endoskeleton is an internal framework found in vertebrates, including humans. It consists of bones and cartilage, providing support, protection, and movement to the body.

Define and explain Scaphopoda:

Scaphopods Are generally rare as fossils. The Scaphopoda, or elephant-tusk shells, have a single, slightly curved high conical shell, open at both ends. They lack gills and eyes, have a mouth equipped with a radula. A foot, surrounded by tentacles similar to that of the bivalves, adapted for burrowing. Carnivorous, feeding on small organisms, foraminiferans. Spend life in quasiinfaunal positions within soft sediment in deeper-water environments. The first scaphopods appeared during the Devonian and apparently had similar lifestyles to living forms such as Dentalium.

Defines trace fossils:

Trace fossils, also known as ichnofossils, are indirect evidence of past organismal activity preserved in sedimentary rocks. They provide valuable information about the behavior, movement, and interactions of ancient organisms. Trace fossils are distinct from body fossils, which are the preserved remains of the actual organisms.

Hutton views about earth:

Early geologists thought the Earth was very young. Scottish scientist James Hutton (1726–1797) chellenged . “no vestige of a beginning,– no prospect of an end”. “ruins of an earlier world” Gaia hypothesis and Modern Concept: 1= Gaia hypothesis: Earth as a living organism in equilibrium with its biosphere. 2= Modern Concept: Earth as a dynamic and changing system, Lithostratigraphy.

Note on biodiversification of GOBE:

The Great Ordovician Biodiversification Event (GOBE) occurred during the Ordovician Period, approximately 485 to 443 million years ago. It was a period of significant biodiversification, marked by the emergence of numerous new species and the development of unique adaptations and ecological strategies. Changes in ocean chemistry, fluctuations in sea levels, and the formation of new habitats likely contributed to this diversification. The GOBE had long-lasting impacts on Earth's ecosystems, shaping complex food webs and influencing the subsequent evolutionary trajectory of life on our planet.

Note on Scala Naturae:

“All organisms were linked in time by a unidirectional ladder, from simplest at the bottom to most complex at the top’’

Scala Naturae, or the "Great Chain of Being," is a concept that originated in ancient Greek and Roman philosophy and was later developed further in medieval and early modern thought. It posits a hierarchical structure of existence with God at the top, followed by various levels representing different forms of life, from minerals to plants, animals, and humans. Each level is believed to be fixed and unchanging, with humans considered superior to other forms of life. This concept has been influential in shaping religious, philosophical, and scientific perspectives on the natural world throughout history.

Principle of natural selection:

Only the organisms best adapted to their environment tend to survive and transmit their genetic characteristics in increasing numbers to succeeding generations while those less adapted tend to be eliminated.

Where oldest rocks are found and their age.

The oldest rocks on Earth are found in Canada and have been dated to be around 3.8 to 4 billion years old. These rocks provide us with valuable information about the early history of our planet. In Australia, there are even some tiny mineral grains that have been found to be even older, dating back to about 4.4 billion years. These ancient rocks and minerals help scientists understand how the Earth formed and evolved over billions of years.

Fossil paddle worm extinct:

paddle worms, an extinct group of marine organisms known as Priapulida, existed during the Cambrian and Ordovician periods around 540 to 440 million years ago. They had a unique body shape with a paddle-like structure and were soft-bodied, which made their fossilization rare. However, fossilized remains of paddle worms have been found in various locations worldwide, offering valuable insights into ancient marine ecosystems and the evolutionary history of these organisms. Although paddle worms are no longer found in present-day oceans, studying their fossils helps us understand past biodiversity and the evolutionary connections between different groups of organisms.

What was the bangiomorpha:

Bangiomorpha One of the oldest multicellular organisms/eukaryote is Bangiomorpha 1.2 Ga. Multicellular and a member of a modern group that engages in sex.

Feature of Mollusca:

phylum Mollusca, which includes animals like snails, clams, and squids, presented in bullet points:

- Soft-bodied animals with a mantle that often secretes a hard shell made of calcium carbonate.
- Most species have a muscular foot used for locomotion.
- Bilateral symmetry, meaning their body can be divided into two symmetrical halves.
- Well-developed organ systems, including a complete digestive system, circulatory system, and nervous system.
- Variety of feeding strategies, including filter feeding, grazing, scavenging, and predation.
- Most mollusks have a radula, a specialized feeding organ with rows of small teeth.
- Complex respiratory systems, with gills used for extracting oxygen from water.
- Open circulatory system, with a heart pumping blood into a series of sinuses.
- Diverse reproductive strategies, including sexual reproduction with separate sexes or hermaphroditism.
- Many mollusks exhibit a larval stage called a trochophore or veliger.

Define Orthogenesis:

Orthogenesis is a now-discredited evolutionary concept suggesting that organisms have an inherent drive towards predetermined, linear, and unidirectional evolutionary change. It posits that species evolve along fixed trajectories independent of natural selection or external factors.

Note on Crustaceans:

Crustaceans are a large and diverse group of arthropods that inhabit marine and freshwater environments. They possess a hard exoskeleton, jointed appendages, and various body forms. Crustaceans, such as crabs, lobsters, and shrimp, play vital roles in aquatic ecosystems as both predators and prey.

Note on Deuterostomes:

Deuterostomes are a major group of animals that include vertebrates (such as fish, mammals, and birds) and some invertebrates (such as echinoderms and hemichordates). Here are some key points about deuterosomes:

- Deuterostomes are characterized by a unique embryonic development pattern known as deuterostomy. During early development, the anus forms before the mouth.
- The group is named after this developmental feature, with "deutero" meaning "second" and "stome" meaning "mouth" in Greek.
- Deuterostomes exhibit bilateral symmetry and possess a true body cavity called a coelom.
- Vertebrates, including humans, are part of the deuterostome lineage. They have a backbone or vertebral column that provides structural support and protects the spinal cord.

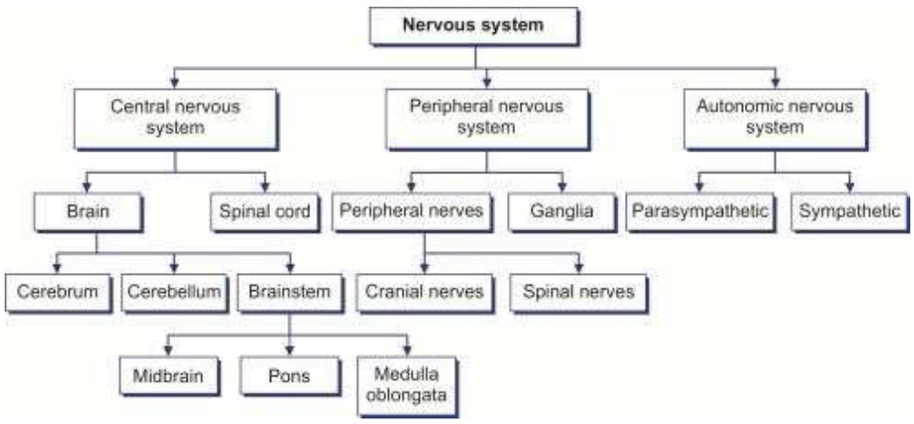
Define Sexual dimorphism:

Sexual dimorphism, in which males and females may show different sizes, and different specialized features (horns, antlers, tail feathers). E.g Sexual dimorphism in ammonites, the Jurassic Kosmoceras. The larger shell was probably the female, the smaller the male. (Courtesy of Jim Kennedy and Peter Skelton.).

Difference between Apomorphy and Pleisiomorphy.

apomorphies are unique traits that evolved more recently and help distinguish one group from others, while pleisiomorphies are shared ancestral traits that are not informative for distinguishing between closely related groups. Both types of character states are important in understanding the evolutionary relationships and patterns of inheritance within a taxonomic group.

Flow chart of CNS:



Write Reptile's and functions:

Reptiles, such as snakes, lizards, turtles, and crocodilians, have scaly skin for protection, lungs for respiration, thermoregulation to adapt to temperature, sexual reproduction, varied diets, diverse locomotion methods, sensory perception, and defense mechanisms for survival in their environments.

Write Lithostratigraphic units:

Lithostratigraphic units are geological units defined based on the type of rock they consist of. They help classify and describe layers of rocks, aiding in understanding geological history and sedimentary environments.

Zone of climate change:

The "Zone of Climate Change" refers to regions or areas experiencing significant shifts in climatic conditions due to anthropogenic factors. These zones are characterized by alterations in temperature, precipitation patterns, sea levels, and ecosystem dynamics, posing challenges to both human and natural systems.

Big five mass extinction events:

The "Big Five" mass extinction events include the Ordovician-Silurian, Late Devonian, Permian-Triassic (Great Dying), Triassic-Jurassic, and Cretaceous-Paleogene (K-T) extinctions. These events resulted in significant loss of biodiversity, with major impacts on Earth's ecosystems and evolutionary history.

Function of paleontology:

Paleontology studies fossil remains to understand the history of life on Earth. It helps reconstruct ancient environments, track evolutionary patterns, study extinct organisms' anatomy and behavior, and provides insights into Earth's past climates, ecosystems, and geological processes.

What is lophophorate:

Lophophorates are a group of small, mostly marine animals characterized by a unique feeding apparatus called a lophophore. The lophophore is a horseshoe-shaped structure with ciliated tentacles used for filter-feeding, and lophophorates include brachiopods, phoronids, and bryozoans.

What is speciation and it's two types?

Speciation often occurs by the establishment of a barrier, and the isolation of part of a previously interbreeding population. “The process of splitting of a population to form two species is speciation ”. 1= The allopatric • (“other homeland”) model 2= The geographic model (based on the establishment of geographic barriers.

The Work of John Phillips:

Formally defined the three great eras separated by extinction events.

1= Paleozoic (“ancient life”) 2= Mesozoic (“middle life”) 3= Cenozoic (“recent life”)

Precise biotic and morphological changes along phylogenetic lineages. Accurate correlation using a wide variety of fossil organisms.

Use of phylogenetics:

Phylogenetics is a field of study that uses genetic data to reconstruct evolutionary relationships between organisms. It helps in understanding the evolutionary history, biodiversity patterns, and genetic divergence among species, aiding in fields such as conservation biology, evolutionary biology, and comparative genomics.

Information and fossil fuels:

Fossil fuels are energy sources derived from the remains of ancient plants and animals, such as coal, oil, and natural gas. They are formed over millions of years through the decomposition and transformation of organic matter under high pressure and temperature in Earth's crust. Fossil fuels have played a significant role in powering human civilization and fueling economic growth due to their high energy content.

Define Cambrian explosion:

The Cambrian explosion generated a range of new body plans during a relatively short time interval. It is refers to a period of rapid diversification and evolution of complex life forms that occurred approximately 541 million years ago during the Cambrian Period. It marked the appearance of diverse animal phyla and the development of complex ecosystems.

Paradigm shifts and its example:

Thomas Kuhn: science shuttles between so-called times of normal science and times of scientific revolution. Scientific revolutions, or paradigm shifts, are when a whole new idea invades an area of science.

An Example of Paradigm shift

The paper by Luis Alvarez and colleagues (1980). The Earth had been hit by a meteorite 65 million years ago, caused the extinction of the dinosaurs and other groups.

Trophoblast:

The trophoblast is a layer of cells that forms the outer layer of the blastocyst, which is an early stage of embryo development. It plays a crucial role in the implantation of the embryo into the uterus and the formation of the placenta.

Deductive reasoning give examples:

Deductive reasoning is a logical process of drawing specific conclusions from general principles or premises. It involves moving from general statements, known as premises, to specific conclusions based on the given information. Here are a few examples of deductive reasoning:

Example: 1: All mammals have hair. Premise 2: A dog is a mammal. Conclusion: Therefore, a dog has hair.

Characters of cephalopod:

Cephalopods, a group of marine mollusks, possess distinctive characters such as tentacles, a beak-like mouth, well-developed eyes, a radula, a mantle and siphon, chromatophores, and intelligence. These features contribute to their unique biology, behavior, and adaptability in the marine environment.

What are the Trilobitomorpha? what are their different body parts?

Trilobitomorpha is a class of extinct arthropods that existed from the Early Cambrian to the Late Permian periods. They had a body divided into three sections: a head with compound eyes and antennae, a thorax with multiple segments bearing legs, and a tail called a pygidium. They were known for their distinctive trilobed exoskeleton.

Characteristics features of pterosaurs:

Pterosaurs were flying reptiles that lived during the Mesozoic Era. They had several characteristic features, including hollow bones, membranous wings supported by an elongated fourth finger, a highly specialized wrist joint, a keeled breastbone for flight muscles, and a long neck and head with sharp teeth.

Three types of amniotic radiated:

Amniotes radiated during the Late Carboniferous, giving rise to three main clades. These are distinguished by the pattern of openings in the side of the skull, especially the temporal openings behind the eye socket. These three skull patterns diagnose the key clades among amniotes which are

1= The anapsid ("no arch") is the primitive state skull pattern, since there are no temporal openings. The Anapsida include various early forms such as Hylonomus, as well as some Permian and Triassic reptiles, and the turtles. The two other skull patterns seen in amniotes are

2= The synapsid ("same arch"), where there is a lower temporal opening. The Synapsida include the "mammal-like reptiles" and the mammals.

3= The diapsid ("two arch") pattern, where there are two temporal openings. The Diapsida includes a number of early groups, as well as the lizards and snakes and the crocodiles, pterosaurs, dinosaurs and birds.